

PSP AWS ADG tip-toe steps

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Pre-requisites

- Verify DG broker health for IBOB + Monolith
- Monolith

```
oracle@pprdrpspdb801.ie.intuit.net:/scripts/adg_test->./chk_broker.sh psp_monolith adgtest
DGMGRL for Linux: Version 12.1.0.2.0 - 64bit Production
```

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Welcome to DGMGRL, type "help" for information.
Connected as SYSDBA.

DGMGRL>

Configuration - PSPQPRD2_DGB

```
Protection Mode: MaxPerformance
Members:
psplpd04 - Primary database
psppqd04 - Physical standby database
psppp003 - Physical standby database
```

Fast-Start Failover: DISABLED

Configuration Status:
SUCCESS (status updated 19 seconds ago)

DGMGRL>

Database - pspqpd04

```
Role:                PHYSICAL STANDBY
Intended State:       APPLY-ON
Transport Lag:        0 seconds (computed 3 seconds ago)
Apply Lag:            0 seconds (computed 3 seconds ago)
Average Apply Rate:   705.00 KByte/s
Real Time Query:      ON
Instance(s):
psppqd04
```

Database Status:
SUCCESS

DGMGRL>

Database - psplpd04

```
Role:                PRIMARY
Intended State:       TRANSPORT-ON
Instance(s):
psplpd04
```

Database Status:
SUCCESS

DGMGRL>

Database - psppp003

```
Role:                PHYSICAL STANDBY
Intended State:       APPLY-ON
Transport Lag:        0 seconds (computed 0 seconds ago)
Apply Lag:            0 seconds (computed 0 seconds ago)
Average Apply Rate:   759.00 KByte/s
Real Time Query:      ON
Instance(s):
psppp003
```

Database Status:
SUCCESS

DGMGRL>

- IBOB

```
oracle@pprdpspdb801.ie.intuit.net:/scripts/adg_test->./chk_broker.sh psp_ibob adgtest
DGMGRL for Linux: Version 12.1.0.2.0 - 64bit Production
```

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Welcome to DGMGRL, type "help" for information.
Connected as SYSDBA.

DGMGRL>

Configuration - dgpsppqd03

```
Protection Mode: MaxPerformance
Members:
psplpd03 - Primary database
psppqd03 - Physical standby database
pspppib3 - Physical standby database
```

Fast-Start Failover: DISABLED

Configuration Status:

SUCCESS (status updated 50 seconds ago)

DGMGRL>

Database - psppqd03

```
Role:                PHYSICAL STANDBY
Intended State:       APPLY-ON
Transport Lag:        0 seconds (computed 0 seconds ago)
Apply Lag:            0 seconds (computed 0 seconds ago)
Average Apply Rate:   73.00 KByte/s
Real Time Query:      ON
Instance(s):
psppqd03
```

Database Status:

SUCCESS

DGMGRL>

Database - psplpd03

```
Role:                PRIMARY
Intended State:       TRANSPORT-ON
Instance(s):
psplpd03
```

Database Status:

SUCCESS

DGMGRL>

Database - pspppib3

```
Role:                PHYSICAL STANDBY
Intended State:       APPLY-ON
Transport Lag:        0 seconds (computed 0 seconds ago)
Apply Lag:            0 seconds (computed 0 seconds ago)
Average Apply Rate:   80.00 KByte/s
Real Time Query:      ON
Instance(s):
pspppib3
```

Database Status:

SUCCESS

DGMGRL>

- AWS - Test and ensure wavefront page alerts are working
- Review IOPS and Instance class settings for monolith and IBOB EC2/volumes.

Tip-toe steps

- Stop Application and make sure there are no connection to database. (Kill if any)

```
Kill existing connections (if still exist)
select 'alter system kill session '||''''||sid||','||serial#||''''||' immediate;' from gv$session where
username in ('PSPADM','PSPAPP') ;
```

- After all connections are gone, trigger db switchover in parallel for IBOB and Monolith as per below command:

```
./db_switchover_test.sh <application_name> <target_site> <encryption_key>
For eg:
./db_switchover_test.sh psp_monolith aws adgtest
./db_switchover_test.sh psp_ibob aws adgtest
```

- below is sample output

```
oracle@pprdpspdb801.ie.intuit.net:/scripts/test->./db_switchover_test.sh adg qdc adgtest

=====
STEP 1: Parse the input and then setup running environment
Switchover is running. Please check log and trace files under /scripts/test/LOG/adg_qdc
oracle@pprdpspdb801.ie.intuit.net:/scripts/test->
```

- Check and verify log (db_switchover.log) under LOG directory for respective switchover as shown above example (/scripts/test/LOG/adg_qdc). it should say "Switchover succeeded" with new primary db.

Post tip-toe steps

- Verify ADG lag and ensure no lag between AWS and IHP
- On IHP

```
./chk_broker.sh <application_name> <encryption_key>
For eg:
./chk_broker.sh psp_monolith adgtest
./chk_broker.sh psp_ibob adgtest
```

- Once extract is ABENDED. Point extract in IHP to Oracle on EC2 in AWS

```
add below to extract parameter file and restart extract
-- Use this login when DB is source is in AWS (oracle on EC2)
USERIDALIAS ggsources_aws
```

Rollback

- Stop Application and make sure there are no connection to database. (Kill if any)

```
Kill existing connections (if still exist)
select 'alter system kill session '||''''||sid||','||serial#||''''||' immediate;' from gv$session where
username in ('PSPADM','PSPAPP') ;
```

- After all connections are gone, trigger db switchover in parallel for IBOB and Monolith as per below command:

```
./db_switchover_test.sh <application_name> <target_site> <encryption_key>
For eg:
--> to switch to QDC
./db_switchover_test.sh psp_monolith qdc adgtest
./db_switchover_test.sh psp_ibob qdc adgtest

--> to switch to LVDC
./db_switchover_test.sh psp_monolith lvdc adgtest
./db_switchover_test.sh psp_ibob lvdc adgtest
```

- below is sample output

```
oracle@pprdpspdb801.ie.intuit.net:/scripts/test->./db_switchover_test.sh adg qdc adgtest

=====
STEP 1: Parse the input and then setup running environment
Switchover is running. Please check log and trace files under /scripts/test/LOG/adg_qdc
oracle@pprdpspdb801.ie.intuit.net:/scripts/test->
```

- Check and verify log (db_switchover.log) under LOG directory for respective switchover as shown above example (/scripts/test/LOG/adg_qdc). it should say "Switchover succeeded" with new primary db.

Post rollback steps

- Verify ADG lag and ensure no lag between AWS and IHP
- On IHP

```
./chk_broker.sh <application_name> <encryption_key>
For eg:
./chk_broker.sh psp_monolith adgtest
./chk_broker.sh psp_ibob adgtest
```

- Once extract is ABENDED. Point extract in IHP back accordingly

```
Uncomment below accordingly in extract parameter file and restart extract
-- Use this login when DB is source is local/LVDC
--USERIDALIAS ggsource

-- Use this login when DB is source is remote/QDC
--USERIDALIAS ggsource_remote
```

Performance tuning

Copy SQL Plan - IHP to AWS

- from cursor cache

```
oracle@pprdpspdb900.ie.intuit.net:/home/oracle/kpopat->./create_spb_aws.sh 627bwsf9sj3g4 803259958 PSP PSP-UW2

Copying plan for sql_id 627bwsf9sj3g4 from cluster# PSP to cluster# PSP-UW2 for plan hash value 803259958

Source DBL:SPM_PSP_LAS
Target DBL:SPM_PSP_UW2
Exporting sql plan for sqlid [627bwsf9sj3g4]
SQL plan export complated!
Importing sql plan for sqlid [627bwsf9sj3g4]
SQL plan import complated!
Export/Import of sql plan completed for sqlid [627bwsf9sj3g4] plan hash [803259958]

PL/SQL procedure successfully completed.
```

- from AWR repository

```
oracle@pprdpspdb900.ie.intuit.net:/home/oracle/kpopat->./create_spb_aws_awr.sh 1wbsa04w60gzt 3307958781 PSP PSP-UW2 4085 4086

Copying plan for sql_id 1wbsa04w60gzt from cluster# PSP to cluster# PSP-UW2 for plan hash value 3307958781

Source DBL:SPM_PSP_LAS
Target DBL:SPM_PSP_UW2
Exporting sql plan for sqlid [1wbsa04w60gzt]
SQL plan export complated!
Importing sql plan for sqlid [1wbsa04w60gzt]
SQL plan import complated!
Export/Import of sql plan completed for sqlid [1wbsa04w60gzt] plan hash [3307958781]

PL/SQL procedure successfully completed.
```